



Apache SINGA: The Deep Learning Model Trainer

Dive into the world of Apache SINGA, a distributed platform for training big deep learning models over large data sets. Discover its architecture, features, and learn how to harness its potential using Docker.

Apache SINGA was developed by the DB System Group of the National University of Singapore in partnership with Zhejiang University's databases group. This system assists in picture identification as well as natural language processing. It embraces many well-known deep learning models, and consists of three main parts: IO Core, Model, and Core. Initiated in 2014, its initial release was on October 8, 2015. Apache Incubator accepted its prototype in March 2015.

Apache SINGA, a distributed deep learning system, provides enhanced IO classes to write, read, encode, and decode files and data. It offers a built-in programming model based on the layer abstraction, which supports a variety of popular deep learning models. This application can be used to train synchronously, asynchronously, or in a combination of both. Hybrid training frameworks can also be modified to achieve good scalability. SINGA delivers different neural net partitioning schemes for training big models. The SINGA project has been released under Apache License 2.

History of Apache SINGA

There has been a huge surge of interest, in both industry and academia, in deep learning. It has provided great accuracy in areas such as multi-modal data analysis and state-of-the-art algorithms. However, many distributed training systems have been proposed to improve runtime performance like Caffe, Purine, Torch, and Google's DistBelief. Through these systems, deep learning models can benefit from deeper structures and larger training data sets.

However, there are a few major challenges in developing a deep learning system. For example, there are a large number of parameters that suffer from a vast amount of communication overhead to synchronise nodes. Therefore, the scalability in terms of training time to reach a certain degree of accuracy is a challenge. Another challenge is that it is non-trivial for programmers to develop and train models with deep and complex model structures. Distributed training further increases the load of programmers, e.g., data and model partitioning, and network communication.

Hence, a distributed deep learning platform was introduced. SINGA is designed with a built-in programming model that supports a variety of popular deep learning models, such as energy models like restricted Boltzmann machines (RBMs), and feed-forward models including convolutional neural networks (CNNs).

SINGA architecture

The SINGA architecture is very flexible and can run asynchronous, synchronous and hybrid training frameworks. Synchronous training enhances the efficiency of one iteration, and asynchronous training boosts the convergence rate. If the user has a fixed cluster size, then SINGA can run a hybrid framework that maximises scalability by trading off between efficiency and the convergence rate. SINGA can also help with different neural net partitioning programs to parallelise the training of large models, either through hybrid partitioning or feature dimension partitioning, such as partitioning on the batch dimension. It has GPU